

CM0859 – MT5009 Type Theory

Natural Deduction

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Preliminaries

Textbook

Pfenning (2023). Lecture Notes on Natural Deduction.

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Conventions

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Preliminaries

Logical constants

$\wedge$	(and)	conjunction
$\vee$	(or)	(inclusive) disjunction
$\supset$	(if —, then —)	implication
$\perp$	(falsity)	bottom, falsum
$\top$	(truth)	top
$\forall x$	(for every x)	universal quantifier
$\exists x$	(there exists a x)	existential quantifier

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Definition

We define negation by $\neg A := A \supset \perp$.

Formalisation of Logical Systems

In relation to the notions of **propositions** and **proofs** the textbook says:

“There is no universal agreement about the proper foundations for these notions.” (p. L2.1)

Formalisation of Logical Systems

Logical System {
 Axiomatic proof theory (Hilbert)
 Structural proof theory (Gentzen) {
 Natural deduction
 Sequent calculus

Formalisation of Logical Systems[†]

Logical System $\left\{ \begin{array}{l} \text{Axiomatic proof theory (Hilbert)} \\ \text{Structural proof theory (Gentzen)} \end{array} \right. \left\{ \begin{array}{l} \text{Natural deduction} \\ \text{Sequent calculus} \end{array} \right.$

[†]Complementary material: Troelstra and Schwichtenberg ([1996] 2000) and Negri and von Plato ([2001] 2008).

Judgements and Propositions

“The cornerstone of Martin-Löf’s foundation of logic is a clear separation of the notions of judgment and proposition.” (p. L2.1)

Judgements and Propositions

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“He [Martin-Löf (1996) and Martin-Löf (1987)] reasons that to judge is to know and that an evident judgment is an object of knowledge. A proof is what makes a judgment evident. In logic, we make particular judgments such as ‘ A is a proposition’ or ‘ A is true’, presupposing in the latter case that A is already known to be a proposition. To know that ‘ A is a proposition’ means to know what counts as a verification of A , whereas to know that ‘ A is true’ means to know how to verify A .” (Pfenning and Davies 2001, p. 512)

Judgements and Propositions

Examples (judgements)

- “ A is a proposition”.
- “The proposition A is true”.
- “The proposition A is false”.
- “The proposition A is true at time t ” (temporal logic).
- “The proposition A is necessarily true” (modal logic).
- “ σ is a type” (programming languages).
- “ M is a combinator/program” (programming languages).
- “The combinator/program M has type σ ” (programming languages).

Judgements and Propositions

From the sharp separation of the notions of judgement and proposition by Martin-Löf, we know that inference rules operate on judgements and logical constants operate on propositions.

Natural Deduction for Propositional Intuitionistic Logic

Introduction

- Propositions: $A, B, C, \dots$
- Judgements:
 - (i) A **prop** (A is a proposition)
 - (ii) A **true** (the proposition A is true)
- Form of the inference rules:

$$\frac{J_1 \quad \dots \quad J_n}{J} \text{ rule name}$$

where J (conclusion) and $J_1, \dots, J_n$ (premises or hypotheses) are all judgements.

- Inference rules: Formation, introduction and elimination rules

Natural Deduction for Propositional Intuitionistic Logic

We shall define inference rules for conjunction, implication, disjunction, truth, falsehood and negation.

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for conjunction

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for conjunction

- Formation rule

$$\frac{A_{\text{prop}} \quad B_{\text{prop}}}{A \wedge B_{\text{prop}}} \wedge\text{F}$$

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for conjunction

- Formation rule

$$\frac{A_{\text{prop}} \quad B_{\text{prop}}}{A \wedge B_{\text{prop}}} \wedge F$$

- Introduction rule

$$\frac{A_{\text{true}} \quad B_{\text{true}}}{A \wedge B_{\text{true}}} \wedge I$$

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for conjunction

- Formation rule

$$\frac{A_{\text{prop}} \quad B_{\text{prop}}}{A \wedge B_{\text{prop}}} \wedge F$$

- Introduction rule

$$\frac{A_{\text{true}} \quad B_{\text{true}}}{A \wedge B_{\text{true}}} \wedge I$$

- Elimination rules

$$\frac{A \wedge B_{\text{true}}}{A_{\text{true}}} \wedge E_1$$

$$\frac{A \wedge B_{\text{true}}}{B_{\text{true}}} \wedge E_2$$

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for implication

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for implication

- Formation rule

$$\frac{A_{\text{prop}} \quad B_{\text{prop}}}{A \supset B_{\text{prop}}} \supset\text{F}$$

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for implication

- Formation rule

$$\frac{A_{\text{prop}} \quad B_{\text{prop}}}{A \supset B_{\text{prop}}} \supset F$$

- Introduction rule

$$\frac{\displaystyle \frac{}{A_{\text{true}}}^u \quad \vdots \quad B_{\text{true}}}{A \supset B_{\text{true}}} \supset I^u$$

In the application of the $\supset I$ rule, we may discharge zero, one, or more occurrences of the assumption.

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for implication

- Formation rule

$$\frac{A_{\text{prop}} \quad B_{\text{prop}}}{A \supset B_{\text{prop}}} \supset F$$

- Introduction rule

$$\frac{}{A_{\text{true}}}^u$$
$$\vdots$$
$$\frac{B_{\text{true}}}{A \supset B_{\text{true}}} \supset I^u$$

In the application of the $\supset I$ rule, we may discharge zero, one, or more occurrences of the assumption.

- Elimination rule

$$\frac{A \supset B_{\text{true}} \quad A_{\text{true}}}{B_{\text{true}}} \supset E$$

Natural Deduction for Propositional Intuitionistic Logic

Observation

For constructing proofs (derivations) in natural deduction we usually proceed by a combination of bottom-up and top-down reasoning as illustrated by the following examples.

Natural Deduction for Propositional Intuitionistic Logic

Example

A derivation where the assumption and the conclusion are the same and the assumption is discharged once. A proof that $A \supset A$ true.

$$\frac{\frac{}{A \text{ true}}^u}{A \supset A \text{ true}} \supset I^u$$

Natural Deduction for Propositional Intuitionistic Logic

Example

A derivation using the same assumption twice. A proof that $(A \wedge B) \supset (B \wedge A)$ true.

$$\frac{\frac{\frac{}{A \wedge B \text{ true}}^u}{B \text{ true}} \wedge E_2 \quad \frac{\frac{}{A \wedge B \text{ true}}^u}{A \text{ true}} \wedge E_1}{B \wedge A \text{ true}} \wedge I \quad \frac{}{(A \wedge B) \supset (B \wedge A) \text{ true}} \supset I^u$$

Natural Deduction for Propositional Intuitionistic Logic

Example

A derivation where there is a vacuous discharge when using the inference rule $\supset I$. A proof that $A \supset (B \supset A)$ true.

$$\frac{\frac{\frac{}{A \text{ true}}^u}{B \supset A \text{ true}} \supset I \text{ (vacuous discharge of } A)}{A \supset (B \supset A) \text{ true}} \supset I^u$$

Natural Deduction for Propositional Intuitionistic Logic

Notation

Let Γ be a set of judgements and let A^{true} be a judgement. The relation $\Gamma \vdash A^{\text{true}}$ means that there is a derivation with conclusion A^{true} from the set of hypotheses Γ .

Natural Deduction for Propositional Intuitionistic Logic

Example

A derivation using the same hypothesis twice. A proof that $A \wedge B \text{ true} \vdash B \wedge A \text{ true}$.

$$\frac{\frac{A \wedge B \text{ true}}{B \text{ true}} \wedge E_2 \quad \frac{A \wedge B \text{ true}}{A \text{ true}} \wedge E_1}{B \wedge A \text{ true}} \wedge I$$

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for disjunction

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for disjunction

- Formation rule

$$\frac{A_{\text{prop}} \quad B_{\text{prop}}}{A \vee B_{\text{prop}}} \vee\text{F}$$

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for disjunction

- Formation rule

$$\frac{A_{\text{prop}} \quad B_{\text{prop}}}{A \vee B_{\text{prop}}} \vee F$$

- Introduction rules

$$\frac{A_{\text{true}}}{A \vee B_{\text{true}}} \vee I_1$$

$$\frac{B_{\text{true}}}{A \vee B_{\text{true}}} \vee I_2$$

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for disjunction

- Formation rule

$$\frac{A_{\text{prop}} \quad B_{\text{prop}}}{A \vee B_{\text{prop}}} \vee\text{F}$$

- Introduction rules

$$\frac{A_{\text{true}}}{A \vee B_{\text{true}}} \vee\text{I}_1$$

$$\frac{B_{\text{true}}}{A \vee B_{\text{true}}} \vee\text{I}_2$$

- Elimination rule

$$\frac{\frac{A_{\text{true}}}{\vdots}^u \quad \frac{B_{\text{true}}}{\vdots}^w}{\frac{A \vee B_{\text{true}} \quad C_{\text{true}}}{C_{\text{true}}} \vee\text{E}^{u,w}}$$

Natural Deduction for Propositional Intuitionistic Logic

Example

A proof that $(A \vee B) \supset (B \vee A)$ true.

$$\frac{\frac{\frac{}{A \vee B \text{ true}}^u}{A \vee B \text{ true}} \quad \frac{\frac{\frac{}{A \text{ true}}^v}{A \text{ true}} \vee I_2}{B \vee A \text{ true}} \quad \frac{\frac{\frac{}{B \text{ true}}^w}{B \text{ true}} \vee I_1}{B \vee A \text{ true}}}{B \vee A \text{ true}} \vee E^{v,w} \quad \supset I^u}{(A \vee B) \supset (B \vee A) \text{ true}}$$

Natural Deduction for Propositional Intuitionistic Logic

Example

A proof that $\{A \vee B \text{ true}, A \supset C \text{ true}, B \supset C \text{ true}\} \vdash C \text{ true}$.

$$\frac{A \vee B \text{ true} \quad \frac{A \supset C \text{ true} \quad \frac{}{A \text{ true}}^u}{C \text{ true}} \supset E \quad \frac{B \supset C \text{ true} \quad \frac{}{B \text{ true}}^w}{C \text{ true}} \supset E}{C \text{ true}} \vee E^{u,w}$$

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for truth

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$$\frac{}{\top_{\text{prop}}} \top F$$

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Inference rules for truth

- Formation rule

$$\frac{}{\top_{\text{prop}}} \top F$$

- Introduction rule

$$\frac{}{\top_{\text{true}}} \top I$$

- Elimination rule

“We have no information if we know $\top_{\text{true}}$, so there is no elimination rule.” (p. L2.2)

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for falsehood

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for falsehood

- Formation rule

$$\frac{}{\bot \text{ prop}} \bot F$$

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for falsehood

- Formation rule

$$\frac{}{\bot \text{ prop}} \bot F$$

- Introduction rule

“ $\bot$ is a proposition that should have no proof! Therefore there are no introduction rules.” (p. L2.10)

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for falsehood

- Formation rule

$$\frac{}{\bot \text{ prop}} \bot F$$

- Introduction rule

“ $\bot$ is a proposition that should have no proof! Therefore there are no introduction rules.” (p. L2.10)

- Elimination rule

$$\frac{\bot \text{ true}}{C \text{ true}} \bot E$$

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for negation

Since we define negation by $\neg A := A \supset \perp$, for it we talk about **derived** inference.

Natural Deduction for Propositional Intuitionistic Logic

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- Formation rule

$$\frac{A_{\text{prop}}}{\neg A_{\text{prop}}} \neg\text{F}$$

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for negation

Since we define negation by $\neg A := A \supset \perp$, for it we talk about **derived** inference.

- Formation rule

$$\frac{A_{\text{prop}}}{\neg A_{\text{prop}}} \neg F$$

- Introduction rule

$$\frac{\frac{}{A_{\text{true}}}^u \quad \vdots \quad \frac{\perp_{\text{true}}}{\neg A_{\text{true}}}}{\neg A_{\text{true}}} \neg I^u$$

Natural Deduction for Propositional Intuitionistic Logic

Inference rules for negation

Since we define negation by $\neg A := A \supset \perp$, for it we talk about **derived** inference.

- Formation rule

$$\frac{A \text{ prop}}{\neg A \text{ prop}} \neg F$$

- Introduction rule

$$\frac{}{A \text{ true}}^u$$
$$\vdots$$
$$\frac{\perp \text{ true}}{\neg A \text{ true}} \neg I^u$$

- Elimination rule

$$\frac{\neg A \text{ true} \quad A \text{ true}}{\perp \text{ true}} \neg E$$

Natural Deduction for Propositional Intuitionistic Logic

Examples

See many more examples in (Carr 2018).

Natural Deduction for Propositional Classical Logic

There are different ways to define a system of natural deduction for classic logic from the previous system of natural deduction for intuitionistic logic. For example, we can add the following inference rule for proof by contradiction (*reductio ad absurdum*).

(See, e.g. (van Dalen [1980] 2013))

$$\frac{\begin{array}{c} \frac{}{\neg A \text{ true}}^u \\ \vdots \\ \frac{\perp \text{ true}}{A \text{ true}} \end{array}}{A \text{ true}} \text{RAA}^u$$

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